

Cron Jobs: Quick Guide

Schedule tasks in Linux: short notes with examples and commands

1. What is a Cron Job?

A **cron job** is a command or script that Linux runs **automatically** at the time you set. A background program called the **cron daemon** checks every minute and runs any job whose time has come. Use it for **repeated tasks**: daily backups, cleanup, reports, updates. The server must be running, but you can log out.

```
sudo systemctl status cron          # Ubuntu/Debian (use crond on Red Hat/CentOS)

Active: active (running)
```

2. Manage Jobs with crontab

Command	Meaning
crontab -l	List your jobs
crontab -e	Edit: create or change jobs (opens an editor)
crontab -r	Remove ALL jobs (no confirmation, be careful!)

Jobs are **user-specific**: each user has their own list.

3. The Time Format: Five Fields

```
* * * * * command
| | | | +--- Day of week (0-7, Sunday = 0 or 7)
| | | +----- Month (1-12)
| | +----- Day of month (1-31)
| +----- Hour (0-23, 24-hour clock)
+----- Minute (0-59)
```

Symbol	Meaning	Example
*	Every value	* * * * * = every minute
,	List	0 0 10,20,30 * * = on 10th, 20th, 30th
-	Range	0 9 * * 1-5 = Mon to Fri at 9 AM
/	Every N	*/15 * * * * = every 15 minutes

Examples to read

Schedule	Meaning
30 14 * * *	Every day at 2:30 PM

Schedule	Meaning
30 14 * * 0	Every Sunday at 2:30 PM
30 14 15 * *	15th of every month at 2:30 PM
0 0 * * *	Every day at midnight
@reboot	Once, every time the server starts
@daily / @weekly / @monthly	Midnight daily / Sunday / 1st of month

Tip: Search **crontab guru** online: type five fields and it explains them in plain English.

4. Example 1 (Step by Step): Run a Script Every Minute

Task: Goal: every minute, add a line with the date and time to a file. Replace **student** with your user name.

Step 1: Create the script

Save in nano with Ctrl+O, Enter, then Ctrl+X.

```
mkdir ~/cron_lab && cd ~/cron_lab
nano create_file.sh

#!/bin/bash
echo "Cron job ran at $(date)" >> /home/student/cron_lab/test_file.txt
```

Step 2: Make it executable and test it by hand

```
chmod +x create_file.sh
./create_file.sh
cat test_file.txt
```

Step 3: Open the crontab and add the job

Use the **full path**. Save and exit. You will see: crontab: installing new crontab.

```
crontab -e

* * * * * /home/student/cron_lab/create_file.sh
```

Step 4: Verify and watch it work

Wait for the next minute. New lines appear (Ctrl+C stops tail).

```
crontab -l
tail -f ~/cron_lab/test_file.txt
```

Step 5: Stop the job when done

Delete the line in the editor (nano: Ctrl+K).

```
crontab -e
```

5. Example 2 (Step by Step): Every Sunday at 2:30 PM

Step 1: Convert to 24-hour time

2:30 PM = hour 14, minute 30. Sunday = 0. Other fields = *.

```
30 14 * * 0
```

Step 2: Add it with your script

The script must have execute permission (chmod +x).

```
chmod +x /tmp/basic.sh
crontab -e

30 14 * * 0 /tmp/basic.sh
```

Step 3: Check

```
crontab -l
```

If crontab opens vi: vi editor keys: press **i** to type, **Esc** then **:wq** + Enter to save, **:q!** to quit without saving.

6. More Ready Examples

Task	Cron line
Daily backup at 1 AM, save log	0 1 * * * /home/student/backup.sh >> /home/student/backup.log 2>&1
Every 15 minutes	*/15 * * * * /usr/bin/df -h / >> /home/student/disk.log
Weekdays at 9 AM	0 9 * * 1-5 /home/student/report.sh
Run a Python program Sunday 2:30 PM	30 14 * * 0 /usr/bin/python3 /home/student/task.py
At every server start	@reboot /home/student/startup.sh

Part meaning: >> file adds output to a log file, and 2>&1 puts error messages in the same file.

7. Troubleshooting and Rules

Problem	Fix
Job does not run	Check cron is running: <code>sudo systemctl status cron</code>
Script does not start	<code>chmod +x script.sh</code> and test it by hand first
Command not found	Use full paths (find with: <code>which command</code>)
% sign in the line	Write <code>\%</code> inside crontab, or put the command in a script

Problem	Fix
Want proof it ran	<code>sudo grep CRON /var/log/syslog</code> (Red Hat: <code>sudo less /var/log/cron</code>)
Wrong time format	Cron refuses to save it. Answer n, then fix the line

8. Cheat Sheet

Line / command	Meaning
<code>crontab -l / -e / -r</code>	List / Edit / Remove all
<code>* * * * * cmd</code>	Every minute
<code>*/5 * * * * cmd</code>	Every 5 minutes
<code>0 * * * * cmd</code>	Every hour
<code>0 0 1 * * cmd</code>	1st of every month, midnight
<code>0 18 * * 1-5 cmd</code>	Mon to Fri at 6 PM

Remember: Always: (1) use full paths, (2) make scripts executable, (3) test by hand first, (4) save output to a log.